

DESIGNED BY QUDUS ELITE LLC

On-Demand Healthcare Staffing & Visit Management Portal

Master Feature Scope Document & System Blueprint

System Architecture:	Qudus Elite LLC
Project Name:	[Unknown - Temporary Title]
Document Version:	v1.0.0 (Production-Ready)
Target Stack:	Laravel API, Flutter Native Mobile, Web Dashboards
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1. Project Overview & Scope

This master feature scope document defines the technical and functional architecture for a high-performance, on-demand healthcare staffing marketplace. Designed by Qudus Elite LLC, the platform bridges the gap between commercial healthcare facilities (such as home care agencies, skilled nursing facilities, and hospitals) and independent, certified healthcare professionals.

The platform enables end-to-end management of clinical patient care cases and shift schedules. It includes a modern web-based administration suite, property-scoped management dashboards for healthcare clients, a low-latency native mobile application for field staff, and a secure backend engine designed to ensure atomicity, scalability, and HIPAA-conscious operations.

Core Ecosystem Architecture

- **Admin Dashboard (Web Suite):** Central operational command center for system owners to control user states, audit documents, manage payments, and resolve disputes.
- **Facility Dashboard (Web Suite):** Dedicated multi-tenant environment for healthcare organizations to dispatch shifts, coordinate client care, and audit electronic visits.
- **Nurse Mobile App (Flutter Native):** Highly optimized cross-platform application (iOS and Android) for field professionals to locate assignments, track earnings, upload compliance data, and verify visits.
- **Backend API Engine:** High-throughput REST API written in Laravel and backed by Redis to manage transactions, queue processing, and data routing.

2. User Roles & Access Hierarchies

The system operates under a strict Role-Based Access Control (RBAC) mechanism to guarantee secure data isolation across tenant boundaries.

2.1 Principal User Personas

- **Super Admin:** Operational root managers with absolute system-wide command over all user accounts, transaction models, database audits, and security controls.
- **Facility Administrator:** The client tenant manager account responsible for setting up corporate workspaces, inviting employees, handling payment details, and posting shifts.
- **Team Members:** Auxiliary facility operators granted custom access parameters by the Facility Administrator to post shifts, view chat channels, or check scheduling records.
- **Nurse / Healthcare Professional:** Verified field workers using the Flutter mobile application to find work, communicate with facilities, and execute shift clock-in/out protocols.

2.2 Field Professional Stratification

The application supports clear technical designations to filter assignment rules and enforce legal clinical compliance:

Designation	Professional Title	Ecosystem Permissions & Core Capabilities
RN	Registered Nurse	Authorized for all visit types including OASIS Assessments, Recertifications, and Supervisory updates.
LVN / LPN	Licensed Practical / Vocational Nurse	Authorized to claim specialized regular shifts, medication administration cases, and standard visits.
CNA	Certified Nursing Assistant	Authorized for primary daily patient care, assistive visits, and general non-assessment shifts.
HHA	Home Health Aide	Authorized for baseline supportive home visits, personal hygiene tasks, and daily non-clinical monitoring.
Therapist	Physical / Occupational / Speech (Future)	Pipeline model designed for rehabilitative home therapy assignments and functional check sheets.
Caregiver	Non-Medical Care Provider (Optional)	Authorized for basic companion services, meal tracking, and general lifestyle home care.

3. Authentication & Session Security Subsystems

A unified authentication layer managing multi-tenant verification across both web profiles and native mobile setups.

3.1 Web Authentication Features

- **Secure Registration & Domain Sorting:** Captures structural details during signup. Enforces business email domain registration for facility clients and validates corporate identities.
- **Role-Based Login Routing:** Dynamically validates security scopes during authentication and routes users straight to their specific control console (Super Admin or Facility Workspace).
- **Multi-Factor Authentication (MFA):** Optional multi-factor validation via SMS or email using OTP challenge blocks powered by Twilio.
- **Standard Security Features:** Implements secure passwords, email validations, forgot/reset workflow patterns, and persistent login states via remember-me cookies.

3.2 Mobile Authentication Features

- **Onboarding & Registration Matrix:** Tailored for field operators to sign up directly through Flutter. Captures basic data fields before opening compliance upload screens.
- **One-Time Password (OTP) Verification:** Supports passwordless mobile access or mandatory step-up safety validation using secure mobile verification channels.
- **Token-Based Session Architecture:** Uses secure Bearer tokens managed locally via Flutter's secure storage layer to handle ongoing API access safely.

3.3 Active Session Management

- **Device Management Control:** Records active security parameters including IP addresses, device operating models, and unique identification strings.
- **Session Revocation Protocols:** Allows users to audit currently connected hardware endpoints and drop permissions on specific devices or clear all connections with a single command.

4. Role & Permission Management (RBAC Engineering)

Permissions are strictly bound to data models via a custom database engine structure. This isolates data records completely across multi-tenant barriers.

Granular Permission Matrix

System Operations	Super Admin	Facility Admin	Team Member	Nurse / Professional
Platform Overrides	Allowed	Blocked	Blocked	Blocked
Create/Edit Cases	Allowed	Allowed	Allowed (Scoped)	Blocked
Approve User Credentials	Allowed	Blocked	Blocked	Blocked
Accept Marketplace Shifts	Blocked	Blocked	Blocked	Allowed (Verified Only)
View Financial Ledgers	Allowed (Global)	Allowed (Tenant)	Custom Toggle	Allowed (Personal Wallet)

5. Facility / Workplace Dashboard

A web workspace tailored for healthcare operations teams. It features a persistent sidebar structure linked to specialized platform tools.

5.1 Persistent Workspace Navigation

- **Dashboard** - Live tracking dashboard showing real-time operational health summaries and immediate alert flags.
- **Cases** - Control center to build, evaluate, review, or copy patient care models and ongoing shift logs.
- **Visit Verification** - High-fidelity monitoring panel featuring live electronic verification charts and GPS tracking streams.
- **Chat** - Instant messaging portal supporting real-time workspace text adjustments and file sharing.
- **Team Members** - Employee management panel to handle profile access parameters and operational security updates.
- **Workplace Requirements** - Policy rule manager to assign and update corporate document compliance metrics.

- **Workplace Information** - Configuration engine for corporate addresses, billing contacts, tax identification details, and operational fields.
- **Upcoming Charges & Invoices** - Detailed transaction histories, downloadable invoices, and upcoming balance metrics.
- **Reports** - Custom query tools tracking workspace completion speeds, shift fill rates, and financial analytics.

5.2 Granular Employee Coordination Engine

Enables Facility Administrators to invite secondary operators using secure database mechanics:

```
CREATE TABLE team_members (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  facility_id BIGINT UNSIGNED NOT NULL,
  full_name VARCHAR(150) NOT NULL,
  email VARCHAR(255) NOT NULL UNIQUE,
  mobile_phone VARCHAR(35) NOT NULL,
  office_phone VARCHAR(35) NULL,
  job_title VARCHAR(100) NOT NULL,
  is_active TINYINT(1) DEFAULT 1,
  permissions_mask JSON NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (facility_id) REFERENCES facility_profiles(id) ON DELETE CASCADE
);
```

5.3 Workplace Requirements & Compliance Manager

A programmatic safety matrix where facility managers configure exact compliance checks that independent workers must satisfy before claiming shifts.

- **System-Wide Verification Library:** Tracks active compliance documents including State Nursing Board Licenses, CPR Certificates, Background Clearances, Work Authorizations, SAM Search Reports, OIG Exclusion Reviews, TB Screenings, and Immunization Profiles.
- **Granular Control Mechanisms:** Allows managers to set document parameters as mandatory or optional, change requirements based on certification categories, and search the system verification rules library easily.

6. Case Management System

The core scheduling framework designed to construct patient profiles, organize clinical visit parameters, and track active shifts safely.

6.1 Complete Case Layout Data Architecture

- **Patient Privacy Identifier:** Automatically masks patient identities with randomized programmatic strings (e.g., **Case-PT-7701**) to protect PHI across public marketplace logs.
- **OASIS Assessment Option Toggle:** A dedicated true/false option flag. Enabling this option requires selecting one of four clinical tracking parameters: Admission, Resumption of Care, Recertification, or Discharge.

- **Spatial Mapping:** Converts input addresses into precise decimal coordinate formats (latitude and longitude) using the Google Maps Geocoding API to enforce geofence controls.
- **Clinical Visit Type Classification:** Links assignments to precise clinical protocols: Admission Visit, Regular Shift, Resumption of Care, Recertification (RN Only), Supervisory Check, or Discharge Protocol.
- **Advanced Specialties Index Matrix:** Multi-select indicators indexing specific skills required for matching: Wound Care, Wound Vac, PICC Line, Dementia Care, Tracheostomy, IV Infusion, TPN, G-Tube Management, Ventilator Support, Pediatrics, Complex Medication Administration, or Catheter Care.

6.2 Shift Scheduling Parameters

- **Financial Configuration Spread:** Explicit definition of `charge_rate` (the structural fee invoiced to the corporate facility) and `pay_rate` (the net earnings distributed to the contract provider).
- **Timing & Pattern Recurrence:** Handles duration parameters, fixed schedule formats, estimated start indicators, and multi-day recurrence arrays mapping out exact days of the week.
- **Operational Directives:** Incorporates urgency levels, worker category rules (RN, LVN, CNA, HHA), and optional patient gender preferences.

7. Shift Marketplace & Anti-Double-Booking Protection

The open marketplace pushes live shift fields to mobile devices. It relies on atomic processing parameters to block duplicate bookings completely.

Critical Database Concurrency Engine

To eliminate double-booking race conditions during high-volume traffic events, the backend uses strict database row isolation. When an acceptance request hits the endpoint, the application triggers a pessimistic transaction lock block using `FOR UPDATE` patterns, isolating the matching case entry immediately until processing concludes.

Marketplace Booking Logic Flow

```
START TRANSACTION;

-- Isolate and lock row instantly to avoid simultaneous access
SELECT status FROM cases
WHERE id = target_case_id
FOR UPDATE;

-- Backend checks if current status variable matches 'open' exactly
-- If validation fails: ROLLBACK instantly and return conflict exception response code 409

UPDATE cases
SET status = 'booked'
WHERE id = target_case_id;

INSERT INTO case_assignments (case_id, nurse_id, booked_at)
VALUES (target_case_id, request_user_id, NOW());

COMMIT;
```

8. Visit Verification Subsystem (EVV Engine)

An electronic visit verification (EVV) tracking system built to ensure regulatory compliance and high field operational reliability.

- **Geofenced Authorization Verification:** Field users execute clock events via mobile. The system calculates distance metrics between live mobile coordinates and target address records using the Haversine mathematical algorithm.
- **Discrepancy Control Mechanics:** If a check-in event records a location variance wider than a 200-meter threshold, the action stops. The interface then forces an override request loop that routes straight to an administrator for manual review.
- **Operational Oversight Tracking:** Grants managers access to complete log files, historical clock trails, exception summaries, and tools to process missed visit reports manually.

9. Nurse Mobile Application (Flutter Client)

A native, cross-platform app designed for field specialists to easily interact with the marketplace and handle their clinical tasks.

- **Detailed Profile Configurations:** Houses professional data records, profile images, specialized clinical experience logs, background details, availability toggles, and customized operational radius limits.
- **Credential Pipeline Interface:** Seamless capture tool for field users to submit updated licenses, credentials, and screening logs. Displays status indicators mapping items as pending, approved, or rejected.
- **Interactive Marketplace Maps:** Advanced filtering features allowing professionals to search and organize shifts by distance parameters, pay boundaries, certification match requirements, and urgency rankings.

- **Ecosystem Utilities:** Features direct push notification reminders, WebSocket message counters, personal digital wallets detailing pending earnings, past invoices, and performance score reviews.

10. Messaging, Payments, & Infrastructure Specifications

10.1 Real-Time Chat Infrastructure

Provides multi-tenant communication channels across WebSockets while maintaining compliance tracking logs. Enables interactions between field staff and facility operators, admins and professionals, and admin teams and facilities. Supports instant typing indicators, delivery statuses, and secure document file uploads via signed URLs.

10.2 Payment Ledger & Stripe Architecture

Integrates custom Stripe Connect infrastructure to automate commercial client card charges and contractor payout processing schedules.

```
CREATE TABLE platform_payouts (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  nurse_id BIGINT UNSIGNED NOT NULL,
  case_id BIGINT UNSIGNED NOT NULL,
  gross_charge DECIMAL(10, 2) NOT NULL,
  net_payout DECIMAL(10, 2) NOT NULL,
  system_commission DECIMAL(10, 2) NOT NULL,
  stripe_transfer_id VARCHAR(255) NOT NULL,
  status ENUM('pending', 'settled', 'failed') DEFAULT 'pending',
  paid_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (nurse_id) REFERENCES users(id) ON DELETE RESTRICT,
  FOREIGN KEY (case_id) REFERENCES cases(id) ON DELETE RESTRICT
);
```

10.3 Comprehensive System Infrastructure

The platform runs on an enterprise cloud infrastructure optimized for reliable performance and scalability:

- **Object Storage**

Service Component	Target Stack Integration	Architectural Purpose & Core Execution Role
Hosting Environment	DigitalOcean Droplets	Scalable virtual machines running optimized Ubuntu environments to manage the API.
Database Instance	Managed MySQL / PostgreSQL	Relational engine handling transactional scheduling records and strict multitenancy data.
Caching & Queues	Managed Redis Cluster	Processes database locks, application session pools, and real-time messaging tasks.
DigitalOcean Spaces	S3-compliant storage for hosting encrypted clinical documents and secure uploads.	
Location Engine	Google Maps API	Handles coordinates geocoding, client mileage checks, and spatial verification rules.

10.4 Security Guard Rails & System Controls

The platform implements rigorous security measures across all layers to ensure complete compliance with sensitive healthcare data guidelines:

- **Data Protection & Cryptography:** Restricts standard internet channels to verified TLS 1.3 setups and encrypts sensitive database fields at rest using AES-256 block tools.
- **Patient Data Anonymization:** Automatically converts sensitive personal variables into randomized public cryptographic strings within public tracking feeds to preserve HIPAA compliance integrity.
- **Application Layer Safeguards:** Employs strict API rate throttling limits, cross-site scripting (XSS) filters, secure JWT parameters, SQL injection blocks, and full system activity logging tracks.